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# Interspecific variation in diving activity among *Anopheles gambiae* Giles, *An. arabiensis* Patton, and *An. funestus* Giles (Diptera: Culicidae) larvae

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ABSTRACT: Anopheline larvae generally inhabit the near-surface of aquatic habitats, but they dive and remain at the bottom of these habitats for some time. This study examined forced and voluntary diving behavior and submergence tolerance in the three major African malaria vectors, *Anopheles gambiae* Giles, *An. arabiensis* Patton, and *An. funestus* Giles. The former two species occur sympatrically in temporal and shallow water bodies, while the latter occurs in permanent deeper water bodies. *Anopheles funestus* was the most tolerant of submergence, but the larvae tended to halt their descent before reaching the bottom by attaching onto a wall. The difference in diving behavior between *An. funestus* and the two species in the *An. gambiae* complex may be an adaptation to their contrasting breeding sites, because the former species must spend considerable energy to surface in its typical breeding sites. Both *An. gambiae* and *An. arabiensis* reached the bottom and crawled along the substrate, but *An. gambiae* voluntarily crawled more often than *An. arabiensis*. The possible importance of asymmetric bottom-feeding between these two sympatric species is discussed. ***Journal of Vector Ecology* 32 (1): 112-117. 2007.**

*Keyword Index*: *Anopheles arabiensis, Anopheles funestus, Anopheles gambiae*, diving behavior, larval ecology.

## INTRODUCTION

Diving behavior of mosquito larvae and pupae may be a predator avoidance response, a means to avoid the mechanical shock of a direct raindrop hit, or a way to deter being washed away by flowing water. Larval bottom-feeding also requires diving locomotion. Vertical activity may be an adaptation for animals to adjust to their environment, although it is regulated physiologically. Mosquito pupal diving behavior varies dramatically among mosquito genera. For example, *Culex pipiens* L. and *Anopheles stephensi* Liston make short-duration, shallow dives, whereas *Aedes aegypti* (L.), *Ae. albopictus* (Skuse), and *Ae. triseriatus* (Say) make longer-duration dives, typically to a depth at which they become neutrally or negatively buoyant (Romoser and Lucas 1999). Briegel (2003) demonstrated that increasing the water depth in larval rearing containers has detrimental effects on larval development in *Ae. aegypti, Cx. pipiens, An. albimanus* Wiedemann, *An. gambiae* Giles, and *An. stephensi.* In particular, optimal larval development was observed for all five species in rearing containers with shallow water (0.5–1 cm). However, when larvae were reared in containers with deep water (>5 cm), no anopheline larvae successfully developed into pupae, although Ae. aegypti larvae were the least affected (Briegel 2003). It is not clear why high mortality occurred only in anopheline larvae reared in containers with deep water. Tuno et al. (2004) demonstrated that forced diving in deep water increased *An. gambiae* larval mortality. Perhaps mosquito larvae exhibit greater vertical locomotion activity in containers of deep water, and this may cause increased larval mortality because of increased energy consumption incurred during diving or other unknown mechanisms. No information yet exists on the variation in the ability of anopheline species to tolerate deep diving and submergence.

Here, we report on a comparative study of larval diving behavior in three African malaria mosquito species: *An. gambiae*, *An. arabiensis* Patton, and *An. funestus* Giles. Anopheline larvae are generally believed to inhabit the near-surface of aquatic habitats (Briegel 2003), but some anopheline species, such as *An. gambiae*, dive and remain at the bottom for some time (Tuno et al. 2004). *Anopheles gambiae* and *An. arabiensis* larvae are frequently found in small, temporary, sunlit, turbid pools of water (Gillies and DeMeillon 1968), and they occur sympatrically throughout much of their breeding range. In comparison, *An. funestus* is often found in large semipermanent or permanent water bodies with emergent plants, and the larvae take much longer to develop into pupae (Gillies and DeMeillon 1968). We were particularly interested in the variability of larval diving behavior and physiological tolerance to submergence between the species exploiting contrasting aquatic environments. Multiple approaches, such as observation of larval behavior in the natural habitats or in the laboratory, are possible. We examined larval behavior in the laboratory; therefore, we did not evaluate variation in larval responses to stimuli, such as emergent grasses and other aquatic insects, that occur in natural habitats. We believed that it was

worthwhile to test if the three species show any difference in their vertical locomotion in the laboratory setting as a first step. We wished to avoid misleading conclusions that could differ depending on what kind of stimuli are applied when we have little information on that point. We also examined how the stage of development affected anopheline larval diving behavior and submergence tolerance. We expected that interspecific variation in diving behavior would occur as adaptations to the environment. However, intraspecific variation in diving behavior between young and old larvae is likely regulated by their level of development, whereas interspecific variation likely represents a counterbalance of adaptation to the environment and species-specific physical–physiological ability. The examination of larval behavior in young and old individuals of multiple species may allow us to determine the effects of adaptation, which will be more obvious in the older, mature larvae. In addition, information on larval vertical movement will provide a better understanding of breeding habitat use by the three major African malaria vectors, which exploit contrasting aquatic environments.

## MATERIALS AND METHODS

### Mosquitoes

We used 2nd, 3rd, and 4th instars of *An. gambiae, An. arabiensis,* and *An. funestus.* These were second-generation larvae of field-collected *An. funestus* and third-generation larvae of the other two species. Females of *An. arabiensis* and *An. funestus* were collected in Ahero, and females of *An. gambiae* were collected in Kakamega, both in western Kenya. These populations were maintained in an insectary at the Vector Biology Control and Research Center, Kenya Medical Research Institute, Kisumu, Kenya. Mosquito larvae were reared in basins (40 cm diameter, 5 cm water depth) and fed with ground yeast tablets (Pharmadass Ltd., Harrow, Middlesex, UK). The basins were exposed to sunlight between 0900 and 1700 h.

All experiments and observations were conducted for the three species together under the same environmental conditions in an outbuilding. The outbuilding was roofed with semi-transparent plastic and had plastic mesh walls; thus, all observations were performed under natural sunlight without artificial lightning.

### Diving depth and mortality

Experiments were conducted in 500 ml graduated cylinders (34 cm height, 5 cm inner diameter) filled with dechlorinated tap water that was kept in a pail (20 liter volume) for 1 day prior to the experiment and changed every day. The diving ability of 2nd and 4th instar larvae was examined at two water depths: 20 and 32 cm. An individual larva in 0.5 ml of water was discharged into a 500 ml graduated cylinder using a 5 ml pipette. In most cases, this discharge provoked a dive. When a larva did not dive, we discharged another 0.5 ml of water from the pipette at the surface; this was repeated a maximum of three times or until the larva began to dive. We recorded the depth reached by each larva and whether it returned to the surface. We also recorded the time between when a larva reached the bottom and when it returned to the water–air interface. The relative density of anopheline larvae is >1.0 (Clements 1999); thus, they sink in mid-water if they are motionless. In addition to vertical movement, larvae were often observed stopping their passive descent by attaching onto a wall of the cylinder. When larvae did not immediately swim up to the surface, the time (in minutes) was recorded between when they reached the deepest point and their return to the water surface. Individuals that did not immediately return to the surface were observed for 6 h after the initial discharge because no larvae were alive after submergence for 6 h in preliminary tests. For each water depth, we observed whether larvae (approximately 100 2nd and 200 4th instars for each of the three species) reached the bottom and if they survived and returned to the water surface. Fifty larvae of each species (150 larvae) were discharged into individual graduated cylinders from 0800 to 1100 h, and mortality checks were made for larvae remaining in the bottom 6 h later. This cycle was repeated twice for the 2nd instar and four times for the 4th instar (January and February 2005) for a total of approximately 1,800 larvae. The water was not changed during the experiments for one group of 50 larvae. The water temperature ranged from 25.3 to 30.0° C.

### Voluntary diving experiment

To examine voluntary diving to the bottom without any food to search and crawling on the substrate in larvae, we placed larvae in a mesh enclosure (20 x 20 x 8 cm) in a basin (40 cm diameter) filled with dechlorinated tap water (Figure 1). Larvae had to crawl through the pliant space between the mesh and the bottom of the basin to escape

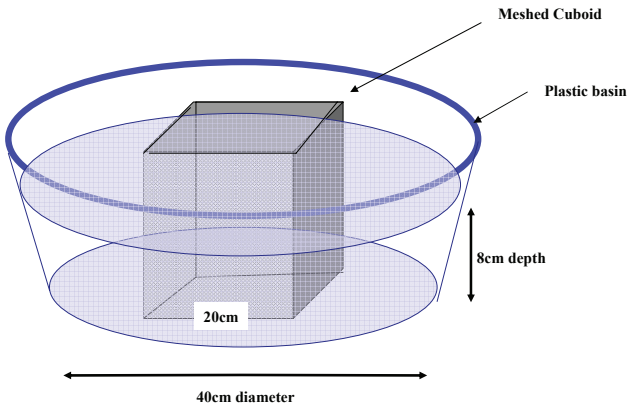



Figure 1. Enclosure apparatus for evaluating the voluntary crawling behavior of anopheline larvae on the bottom substrate. Fifty 4th instar larvae were placed in an enclosure (20 x 20 x 8 cm). The enclosure was covered by fine 0.1-mm mesh, except on the bottom. Mosquito larvae inside the enclosure could crawl out through the pliant mesh edge. The number of larvae remaining inside the enclosure was counted after 1 h of submergence.

from the enclosure. For each species, we gently released 50 4th instar larvae into an enclosure. The number of larvae remaining inside the enclosure was counted after 1 h. This experiment was conducted simultaneously for *An. gambiae*, *An. arabiensis*, and *An. funestus* in different enclosures under the same environmental conditions. The experiment was repeated seven times. In total, we used 1,050 4th instar larvae (three species x seven replicates x 50 individuals). The water temperature ranged between 24.1 and 29.8° C during the trials.

**Submergence tolerance**

To determine the maximum tolerance of mosquito larvae at the bottom of the basins, we placed 20–25 larvae in a 30 ml plastic vial covered with mesh. The plastic vial was sunk in 20 cm of water, to the bottom of a pail filled with 10 liters of dechlorinated tap water. Larval mortality was determined every 2 h for a period of 10 h. As a control, the mortality of larvae in a plastic vial kept at the surface of the pail was also monitored. Two replicates of 2nd, 3rd, and 4th instar larvae of each species were used for each treatment. In the second replicate, the period of observing larval mortality was extended to 12 h. The water temperature varied from 19.6 to 25.3° C.

**Data analysis**

The among-species and among-age variations in the proportion of larvae that reached the bottom of a water cylinder were analyzed using contingency table analysis. If there was significant variation, we performed pairwise comparisons and applied the sequential Bonferroni correction. Analysis of variance (ANOVA) was used to determine the effects of larval age and species on diving duration followed by Tukey–Kramer honestly significant difference (HSD) tests for pairwise comparisons. Diving durations were log-transformed to normalize their distributions. The data for shallow and deep water columns were analyzed separately. One-way ANOVA was conducted to determine the among-species variation in the proportion of larvae escaping from the enclosure followed by Tukey–Kramer HSD tests. Proportional values were arcsine-square root transformed. Survival of larvae at a depth of 20 cm for 10 to 12 h was analyzed using Kaplan–Meier product limit survival analyses. Statistical analyses were performed using JMP version 5.0 (SAS Institute 1994).

## RESULTS

**Diving depth and mortality**

No larval mortality was observed in the 20 cm water column (Table 1). In contrast, 3% of 2nd instar *An. gambiae* and 2–3% of 4th instar *An. gambiae* and *An. arabiensis* drowned from diving in the 32 cm water column, although they attempted to ascend to the surface. These individuals were unable to sustain their activity to reach the surface (Table 1). Contingency table analyses showed that a higher proportion of 4th than 2nd instar larvae reached the bottom in *An. gambiae* (20 cm: $\chi^2 = 8.2$, df = 1, $P < 0.05$; 32 cm: $\chi^2 = 14.9$, df = 1, $P < 0.001$) and *An. arabiensis* (20 cm: $\chi^2 = 88.2$, df = 1, $P < 0.001$; 32 cm: $\chi^2 = 30.4$, df = 1, $P < 0.001$), but not in *An. funestus*. A higher proportion of 2nd instars reached the bottom in *An. gambiae* than in the other two species in 20 cm of water ($\chi^2 = 52.5$, df = 2, $P < 0.001$), whereas there was no difference among the species in 32 cm of water ($\chi^2 = 2.2$, df = 2, $P > 0.05$). A higher proportion of 4th instars of both *An. gambiae* and *An. arabiensis* reached the bottom than *An. funestus* in both 20 cm ($\chi^2 = 96.8$, df = 2, $P < 0.001$) and 32 cm ($\chi^2 = 32.7$, df = 2, $P < 0.001$) of water.

To measure the duration of their dive, another 50 larvae of each species were tested to allow a more powerful

Table 1. Proportion of 2nd and 4th instar larvae of *An. arabiensis, An. gambiae,* and *An. funestus* that dived to the bottom in a water column and mortality rate.

| | | Water column depth | | | | | |
|---|---|---|---|---|---|---|---|
| | | 20 cm | | | 32 cm | | |
| Age | Species | N | Proportion reaching bottom* | Mortality rate | N | Proportion reaching bottom* | Mortality rate |
| 2nd instar | *An. arabiensis* | 100 | 0.01 a | 0.00 | 100 | 0.04 a | 0.00 |
| | *An. gambiae* | 100 | 0.29 b | 0.00 | 101 | 0.06 a | 0.03 |
| | *An. funestus* | 100 | 0.02 a | 0.00 | 100 | 0.02 a | 0.00 |
| 4th instar | *An. arabiensis* | 200 | 0.48 a | 0.00 | 200 | 0.29 a | 0.03 |
| | *An. gambiae* | 200 | 0.46 a | 0.00 | 200 | 0.23 a | 0.02 |
| | *An. funestus* | 200 | 0.09 b | 0.00 | 198 | 0.08 b | 0.00 |

* Letters following numbers indicate the results of pairwise contingency table analysis.
Values with the same letter in a column were not statistically significant at $P < 0.05$ for the same age larvae.

Table 2. Duration (in minutes) of 2nd and 4th instar larvae of *An. arabiensis*, *An. gambiae*, and *An. funestus* remaining under water surface after reaching the deepest point.

| | | Water column depth | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 20 cm | | | | | 32 cm | | | | |
| Age | Species | N | Max | Mean | SD | Tukey-Kramer HSD* | N | Max | Mean | SD | Tukey-Kramer HSD * |
| 2nd instar | *An. arabiensis* | 8 | 5 | 2.0 | 1.4 | c | | | | | |
| | *An. gambiae* | 18 | 15 | 9.0 | 3.5 | a | | | | | |
| | *An. funestus* | 12 | 30 | 3.8 | 8.3 | c | | | | | |
| 4th instar | *An. arabiensis* | 50 | 45 | 7.7 | 8.8 | ab | 59 | 160 | 7.9 | 21.7 | a |
| | *An. gambiae* | 9 | 30 | 5.1 | 6.0 | bc | 37 | 138 | 31.4 | 47.2 | b |
| | *An. funestus* | 45 | 30 | 9.4 | 8.8 | ab | 13 | 180 | 59.5 | 70.9 | c |

*Values with the same letter in a column were not statistically significant at $P < 0.05$ for the same age larvae.

statistical analysis. Times that were less than 1 min were not recorded. Table 2 shows the results. Second instars of *An. arabiensis, An. gambiae*, and *An. funestus* spent 2.0, 9.0, and 3.8 min on average, respectively, and a maximum of 5, 15, and 30 min on dives in the 20-cm water column. In the 32-cm water column, most individuals did not remain submerged for longer than 1 min. Fourth instars of *An. arabiensis, An. gambiae*, and *An. funestus* spent 7.7, 5.1, and 9.4 min on average, respectively, and a maximum of 45, 30, and 30 min on dives in the 20-cm water column. In the 32-cm water column, the respective species spent 7.9, 31.4, and 59.5 min on average, and a maximum of 160, 138, and 180 min on dives (Table 2). ANOVA was used to examine the main and interaction effects of age and species on log-transformed diving duration in each water depth. In the 20-cm water column, all factors had significant effects on larval diving duration. Second instars had shorter dives than 4th instars in *An. arabiensis* and *An. funestus* (Table 2), whereas 2nd instars had longer dives than 4th instars in *An. gambiae* (Table 2). In 32 cm water columns, only data for 4th instars were available. *Anopheles funestus* dove the longest, followed by *An. gambiae*, and *An. arabiensis* (Tukey–Kramer HSD, $P < 0.05$).

**Voluntary diving experiment**

Table 3 shows the mean proportion of the number of individuals that remained in the enclosure and results of the Tukey–Kramer HSD tests. Fewest *An. gambiae* remained in the enclosure (55%), followed by *An. arabiensis* (81%) and *An. funestus*, of which all individuals remained in the enclosure (Table 3).

**Submergence tolerance**

When larvae were enclosed in 30 ml tubes covered with mesh that enabled water exchange, but no air exchange, we observed larval mortality 8 h after enclosure. All *An. gambiae*, *An. arabiensis,* and 2nd instars of *An. funestus* were dead after 12 h, whereas more than half of the 3rd and 4th instars of *An. funestus* survived (Table 4). No mortality was observed in the control treatment in which larvae were maintained in the same manner (but were in water with air exchange). Survival analyses indicated no significant difference among larval ages ($\chi^2 = 0.95$, df = 2, $P = 0.62$), but significant differences among the three species at each age ($\chi^2 = 18.63$, 90.82, and 62.09 for 2nd, 3rd, and 4th instars, df = 2, $P < 0.0001$ in all tests). *Anopheles funestus* larvae survived longer than those of *An. gambiae* and *An. arabiensis* (Table 4).

## DISCUSSION

Substantial differences were found in anopheline mosquito larval diving behavior between *An. funestus*

Table 3. Proportion of anopheline 4th instar larvae remaining within the enclosure 1 h after introduction.

| Species | Number of trial | Mean | SD | Tukey-Kramer HSD test* |
|---|---|---|---|---|
| *An. arabiensis* | 7 | 0.81 | 0.10 | A |
| *An. gambiae* | 7 | 0.55 | 0.18 | B |
| *An. funestus* | 7 | 1.00 | 0.00 | C |

*Values with the same letter in a column were not statistically significant at $P < 0.05$.

Table 4. Survival rates of 2nd, 3rd, and 4th instar anopheline larvae that were forced to submerge for 10 or 12 h.

| Age | Species | Number of trials* | Survival time (hour) | | |
|---|---|---|---|---|---|
| | | | Mean | SE | Wilcoxon Test** |
| 2nd instar | *An. arabiensis* | 2 | 10.35 | 0.16 | a |
| | *An. gambiae* | 2 | 10.20 | 0.10 | a |
| | *An. funestus* | 2 | 9.35 | 0.24 | b |
| 3rd instar | *An. arabiensis* | 2 | 10.55 | 0.18 | a |
| | *An. gambiae* | 2 | 8.52 | 0.15 | b |
| | *An. funestus* | 2 | 11.60 | 0.20 | c |
| 4th instar | *An. arabiensis* | 2 | 9.00 | 0.19 | a |
| | *An. gambiae* | 2 | 9.10 | 0.16 | a |
| | *An. funestus* | 2 | 11.50 | 0.18 | b |

* One trial comprised 20 or 25 larvae.
**Values with the same letter in a column were not statistically significant at $P < 0.05$.

and the two species in the *An. gambiae* complex and among developmental stages. We observed an age effect in diving behavior when older instars reached the bottom of the water column, but there was no general age effect in submergence tolerance. Diving ability is a combination of vertical locomotion ability and submergence tolerance. The results suggest that submergence tolerance is achieved earlier in larval development, whereas vertical locomotion ability is acquired later.

Among-species differences in diving behavior may relate to characteristics of the breeding sites. Forced diving caused mortality in *An. gambiae* and *An. arabiensis* larvae but did not affect *An. funestus* in deeper water columns, possibly because *An. funestus* larvae made significantly fewer dives to the bottom. Fourth instars of *An. funestus* decreased the extent of their descent in the water compared to the two species of the *An. gambiae* complex by adhering to the column wall. The relative density of anopheline larvae is >1.0 (Romoser 1975, Clements 1999), and thus, mosquito larvae can sink passively. As a consequence, larvae that dive deeper will expend more energy swimming back to the water surface. The tendency to stop the passive descent observed in *An. funestus* appears to be an adaptive response because the species inhabits deep water and larvae would expend considerable energy to return to the water surface if they fell to the bottom. This species was also the most tolerant of prolonged submergence, which would enable larvae to save energy by avoiding frequent surfacing to breathe.

The developmental stage also affected diving behavior, except in *An. funestus*. Both 2nd and 4th instars of *An. funestus* prevented their passive descent to the bottom whereas 4th instars of *An. gambiae* and *An. arabiensis* descended passively to the bottom more often than 2nd instars. This asymmetric age effect implies that diving to the bottom is an adaptation only for species that inhabit shallow water. Diving duration was not necessarily consistent with submergence tolerance, indicating that it was not simply regulated by submergence tolerance.

The submergence tolerance tests imply that *An. funestus*, which is able to interrupt passive descent to the bottom, was the most tolerant of long submergence. Thus, *An. funestus* may avoid longer swimming distances by remaining closer to the water–air interface and avoid frequent swimming by making longer submergences.

*Anopheles gambiae* and *An. arabiensis* showed similar proportions of larvae that reached the bottom and similar mortality caused by forced diving and submergence, indicating that there is no difference in their physical diving ability. However, more *An. gambiae* escaped the enclosure, which suggests that *An. gambiae* voluntarily crawl on the substrate surface more than *An. arabiensis*. This difference may be important in maintaining their sympatric distribution in nature in terms of resource partitioning (Toda et al. 1999, Takahashi et al. 2005). Shallow diving does not cause immediate mortality but may affect larval development. Because vertical locomotion requires energy consumption, there must be some compensation for voluntary diving by larvae. These two species have often been observed

feeding on the bottom substrate (N. Tuno, unpublished data). Experiments to evaluate the importance of feeding in proximity to the bottom in these two species showed that the availability of such feeding affected both species but *An. gambiae* more than *An. arabiensis* (N. Tuno, unpublished data). It is likely that more food items, such as algae, are available on the bottom substrate in shallower water, where light can penetrate to promote photosynthesis, than in deeper water. The importance of algae for larval feeding in *An. gambiae* has been discussed (Gimnig et al. 2001, Tuno et al. 2005, Tuno et al. 2006). Thus, feeding in proximity to the bottom may compensate for the larval consumption of energy in vertical locomotion. Algae may also act as a supplier of dissolved oxygen in larval habitats as a by-product of photosynthesis. Effects of water temperature and amount of dissolved oxygen on larval survival during their submergence have not been examined to our knowledge and require further study under appropriate experimental conditions.

In conclusion, the diving activity of mosquito larvae is not regulated simply by their submergence tolerance. We suggest that the differences in diving behavior between *An. funestus* and the two species in the *An. gambiae* complex are adaptations to specific characteristics of their breeding habitats. Differences between the sympatric species *An. gambiae* and *An. arabiensis* were not obvious, except that *An. gambiae* crawled along the bottom substrate more often. This difference may be related to food resource partitioning and requires further study.

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